

Day 4 – Common Cyber Attacks

Aaj hum har attack ko **deep level** par samjhenge:

- ✓ Attack kya hota hai
- ✓ Kaise kaam karta hai (step-by-step)
- ✓ Real life example
- ✓ Technical angle
- ✓ Prevention ka exact logic

Ye notes beginner se interview level tak helpful rahenge 🔥

1 Phishing Attack (Sabse Common Attack)

Basic Definition

Phishing ek **social engineering attack** hai.

Attacker victim ko trick karta hai taaki wo khud hi:

- Password de
- OTP de
- Bank details de
- Login kare fake website par

Yaha hacking tool nahi — **human psychology** use hoti hai.

Attack Ka Psychology

Phishing 3 cheezon par kaam karta hai:

1. Fear 😨 – “Account block ho jayega”
 2. Urgency ⏰ – “Within 1 hour verify karo”
 3. Authority 👮 – “Bank / Government / Police”
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Step-by-Step Kaise Kaam Karta Hai?

1. Attacker fake email ya SMS bhejta hai
 2. Link fake website par le jata hai
 3. User login karta hai
 4. Credentials attacker ke server par store ho jate hain
 5. Real account hack ho jata hai
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Types of Phishing

1. Spear Phishing

Specific employee ko target karna.

Example:

Company HR ko fake invoice bhejna.

2. Whaling

CEO ya CFO ko target karna.

3. Vishing

Fake bank call.

4. Smishing

SMS phishing.



Prevention Logic

- ✓ URL carefully dekho
- ✓ Domain spelling check karo
- ✓ MFA enable karo
- ✓ Employees ko training do



Man-in-the-Middle (MITM) Attack



Basic Concept

Attacker 2 logon ke beech me baith jata hai.

User ↔ Attacker ↔ Website

Victim ko lagta hai direct website se connect hai.



Real Example

Public Wi-Fi par:

Attacker fake Wi-Fi bana deta hai:

"Free_Airport_WiFi"

Tum connect karte ho
wo tumhara traffic read kar sakta hai.



Technical Methods

1. Packet Sniffing

Network packets capture karna (Wireshark jaisa tool)

2. Session Hijacking

Login session cookie steal karna

Prevention

- ✓ HTTPS use karo
- ✓ SSL/TLS encryption
- ✓ Public Wi-Fi me VPN
- ✓ HSTS enable karo

3 Malware Attack

Malware = Malicious Software

System ko:

- Damage
- Control
- Spy
- Destroy

karne ke liye banaya gaya software.

Types Deep Explanation

Virus

File ke sath attach hota hai

User execute kare tab spread hota hai.

Worm

Network ke through automatically spread hota hai.

Trojan

Software jaisa dikhe
andar malicious code.

Example:

Fake cracked software.

Spyware

User ke:

- Keystrokes
- Screenshots
- Browsing data

collect karta hai.

Prevention

- ✓ Antivirus
 - ✓ Software updates
 - ✓ Email attachments careful
 - ✓ Firewall
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4 Ransomware

Advanced Malware

Ye data encrypt karta hai.

Without decryption key → file open nahi hoti.

Attack Flow

1. Entry (Phishing ya vulnerability)
 2. System access
 3. Files encrypt
 4. Ransom note show
 5. Bitcoin demand
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Real Impact

Hospitals

Banks

Companies

Business shutdown ho sakta hai.

Prevention Strategy

- ✓ 3-2-1 Backup rule
- ✓ Offline backup
- ✓ Patch management
- ✓ Network segmentation

5 Brute Force Attack

Simple Concept

Har possible password try karna.

Example:

aaaa

aaab

aaac

Automated tool use hota hai.

Technical Detail

Time depends on:

- ✓ Password length
 - ✓ Character complexity
 - ✓ System lockout
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Prevention

- ✓ Long passwords (12+ chars)
 - ✓ Account lockout
 - ✓ CAPTCHA
 - ✓ MFA
-

6 Dictionary Attack

Smart Brute Force

Common password list use karta hai.

password

admin

qwerty

123456

Difference from Brute Force

Brute Force → All combinations

Dictionary → Common passwords only

Faster hota hai.

Prevention

- ✓ Common passwords avoid
 - ✓ Password manager
 - ✓ MFA
-

7 DoS (Denial of Service)

Concept

Server ko overload kar dena.

Legitimate users access nahi kar pate.

Kaise Hota Hai?

1 attacker

1 system

continuous traffic send karta hai.

Server resources:

- CPU
- RAM
- Bandwidth

exhaust ho jate hain.

Prevention

- ✓ Rate limiting
 - ✓ Firewall
 - ✓ IDS/IPS
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8 DDoS

Advanced DoS

Multiple infected machines (Botnet) use hote hain.

Botnet Kya Hota Hai?

Malware infected devices ka network

malware infected devices as network.

Thousands of systems ek sath attack karte hain.

Why Dangerous?

- ✓ Multiple IP
- ✓ Hard to block
- ✓ Massive traffic

Prevention

- ✓ Cloud protection
- ✓ CDN
- ✓ WAF
- ✓ Traffic filtering

9 SQL Injection

Web Application Attack

Attacker malicious SQL query input field me dalta hai.

Agar validation nahi hai →
Database compromise.

Technical Example

Normal Query:

```
SELECT * FROM users WHERE username='admin' AND password='123';
```

Malicious Input:

```
' OR '1'='1
```

Result:

Login bypass.

Prevention

- ✓ Prepared statements
 - ✓ Input validation
 - ✓ Least privilege
 - ✓ Regular security testing
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Attack Categories Samjho

Category	Attack
Social Engineering	Phishing
Network	MITM
Malware	Virus, Worm, Trojan
Authentication	Brute Force, Dictionary
Availability	DoS, DDoS
Web Application	SQL Injection

Real World Reality

Most common attack today:

1. Phishing
 2. Ransomware
 3. DDoS
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Important Interview Questions

- ✓ Phishing vs Spear Phishing difference?
 - ✓ DoS vs DDoS difference?
 - ✓ Brute Force vs Dictionary difference?
 - ✓ How SQL Injection works?
-

Ultimate Security Formula

Human Awareness

+

Strong Authentication

+

Regular Updates

+

Backups

Strong Cyber Security



Final Thought

Cyber attacks 3 cheezon ko target karte hain:

1. Human
2. Weak passwords
3. Unpatched systems

Agar ye secure hain →

90% attacks fail ho jate hain.